AI HACKATHON'26

PROBLEM STATEMENT

TECHVERSE

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AGENTIC AI HEALTHCARE CHALLENGE 2025

Build Autonomous Clinical Intelligence That Saves Lives

1. THE PROBLEM (2 Minutes to Read This)

The Story of Fatima (Lahore, 2:00 AM): Fatima wakes with crushing chest pain. She calls a helpline. An untrained operator says, "Come in if you think it's serious." She waits until 6 AM. Doctors diagnose acute appendicitis hours before rupture.

The Story of Dr. Ahmed (Rural Sindh, 9:00 AM): Dr. Ahmed sees 100 patients daily (3 minutes each). A farmer says, "I'm tired and thirsty." Dr. Ahmed prescribes vitamins. Three weeks later: diabetic ketoacidosis, blood glucose 450 mg/dL.

The Gap

- **1 doctor per 1,000 people** in Pakistan.
- **2–3 minute consultations** on average.
- **50–70% of medical errors** are caused by communication failures.
- **Language barriers** (English/Urdu vs. Punjabi/Pashto/Sindhi).

> **The Vision:** What if AI could conduct thorough 15-minute interviews in the patient's language, identify red flags, and hand doctors complete, structured information before they even enter the room?

Detail	Information
Duration	48 Hours

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Prize Pool	45k PKR
Team Size	2–4 members
Deadline	11th Jan, 2026

2. WHAT SUCCESS LOOKS LIKE [14]

Case Study 1: Multi-Step Clinical Interview Agent

The Problem: Short consultations lead to missed metabolic crises (e.g., the Diabetic Farmer).

Feature	Without Agentic AI (Current Reality)	With Your Agentic AI System
Interaction	2-minute rushed consultation.	15-minute thorough, adaptive interview.
Language	Potential language barrier.	Conducted in native Punjabi/Urdu.
Result	Undiagnosed diabetes; Emergency 3 weeks later.	Same-day diagnosis and immediate treatment.

Agent Interview Flow (Example: Punjabi)

- **AI:** "میں سمجھتا ہوں۔ آپ کو کب سے تھکاوٹ محسوس ہو رہی ہے؟" (How long have you been feeling tired?) [23232323]

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- **Patient:** "تقریباً 2-3 مہینے" (About 2-3 months)
- **[REASONING AGENT]:** Chronic fatigue → rule out metabolic causes.
- **AI:** "کیا آپ کو بہت زیادہ پیاس لگتی ہے؟" (Do you feel very thirsty?)
- **Patient:** "ہاں، بہت زیادہ" (Yes, very much)
- **[REASONING AGENT]:** Fatigue + polydipsia → diabetes screening triggered.

Generated Clinical Summary (SOAP Format)

- **SUBJECTIVE:** Chief Complaint: Chronic fatigue (2-3 months). History: Progressive tiredness, polydipsia, 3-4x nocturia, unintentional weight loss (~5 kg). Family History: Father with Type 2 Diabetes.
- **ASSESSMENT:** HIGH PRIORITY: Suspected Type 2 Diabetes Mellitus. Classic triad present.
- **PLAN:** 1. URGENT: Random blood glucose or HbA1c. 2. Rule out: Diabetic ketoacidosis if glucose > 250 mg/dL.

Case Study 2: Real-Time Emergency Triage Orchestrator

The Problem: Delayed response to life-threatening symptoms (e.g., 4-hour delay for chest pain).

Agent Challenges:

- Assign WHO Emergency Triage (IMMEDIATE / VERY URGENT / URGENT / STANDARD).
- Generate action plan: "Call 1122 NOW" vs. "Visit ER within 2 hours."
- Work via WhatsApp (text-only, ≤ 50kb, 2G compatible).

Red Flag Detection Required:

- Chest pain + radiation + diaphoresis → MI.

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- Sudden severe headache + neck stiffness → SAH.
- Difficulty breathing + leg swelling → PE.

Case Study 3: Medication Interaction Intelligence System

The Problem: Dangerous polypharmacy complications (e.g., warfarin + aspirin + NSAID interaction).

Agent Must:

- Ingest patient's complete medication list.
- Build a dynamic interaction graph (drug-drug, drug-disease, drug-food).
- Classify severity: CONTRAINDICATED / MAJOR / MODERATE / MINOR.

Test Scenario:

- **Patient:** 68F, CKD Stage 3, Diabetes.
- **New Rx:** Ibuprofen 400mg.
- **Agent Alert:** Flag Ibuprofen + CKD (nephrotoxic risk) and Ibuprofen + Aspirin (MAJOR GI bleed risk).

Case Study 4: Mental Health Crisis Detection & Safety Planning

The Problem: 1 psychiatrist per 10,000 people; suicides often go undetected due to stigma.

Agent Constraints:

- 100% detection of active suicidal ideation (zero false negatives).
- Use trauma-informed, culturally sensitive language.
- Build a personalized safety plan (contacts, coping strategies).
- Immediate human escalation for active crises.

Case Study 5: Cross-Lingual Medical Knowledge Translator

The Problem: Rural patients often do not understand complex English medical terminology.

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Agent Translation Levels:

- **Level 1:** 5th-grade literacy (analogies).
- **Level 2:** High-school graduate.
- **Level 3:** College-educated.
- **Contextual adaptation:** e.g., Diabetes → "شوگر کی بیماری".

Case Study 6: Clinical Decision Support - Lab Report Analyzer

The Problem: Patients wait weeks to have 15-page lab reports interpreted by a doctor.

Agent Must:

- Parse lab reports (PDF/Image) into structured data.
- Compare values against reference ranges (age/gender/pregnancy adjusted).
- Identify critical values (e.g., K+ > 6.5, Glucose > 400).
- **NEVER diagnose** (flag abnormalities + explain significance only).

Case Study 7: Predictive Hospital Readmission Prevention

The Problem: Patients are readmitted shortly after discharge due to missed early deterioration.

Daily Check-in Tasks:

- Automated tracking: "Any swelling? Weight today? Breathlessness?"
- Detect trajectories: e.g., +2kg weight gain in 48 hours = fluid retention.
- Predict readmission risk 48–72 hours before it becomes an emergency.

3. YOUR MISSION

Build an agentic AI system that:

1. Conducts autonomous patient interviews (text or voice).
2. Asks adaptive follow-up questions based on responses.

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3. Identifies potential emergency red flags.
4. Generates structured clinical summaries for doctors

System Boundaries

- **What This System IS:** A clinical decision-support tool; an intelligent medical intake assistant; a triage support system.

- **What This System IS NOT:** A diagnostic system; a treatment advisor; a replacement for doctors.

4. WHAT "AGENTIC" MEANS (vs. Regular Chatbots)

- **Regular Chatbot:** User says "I have a headache." Bot says "Take paracetamol and rest." (End of conversation) .

- **Agentic AI System:** User says "I have a headache." AI reasons about sudden onset, severity, and red flags. It asks about "Thunderclap" onset and triggers an Emergency Alert for potential Subarachnoid Hemorrhage.

5. MINIMUM VIABLE PRODUCT (MVP)

- **Bronze Tier :** Handles ONE clinical scenario; 5–10 adaptive questions; Generates SOAP summary.
- **Silver Tier :** Handles 3 scenarios; Multi-agent architecture; Red-flag detection logic.
- **Gold Tier :** Voice/Multilingual support; Real-time explainability; Integration-ready (WhatsApp/API).

6. DATASETS PROVIDED (FREE)

- **Official Challenge Dataset:** 50 synthetic clinical cases validated by Pakistani doctors.
- **MedQA:** GitHub Link.
- **PubMedQA:** GitHub Link.
- **MedMCQA:** MedMCQA Website.

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Format Preview (JSON)

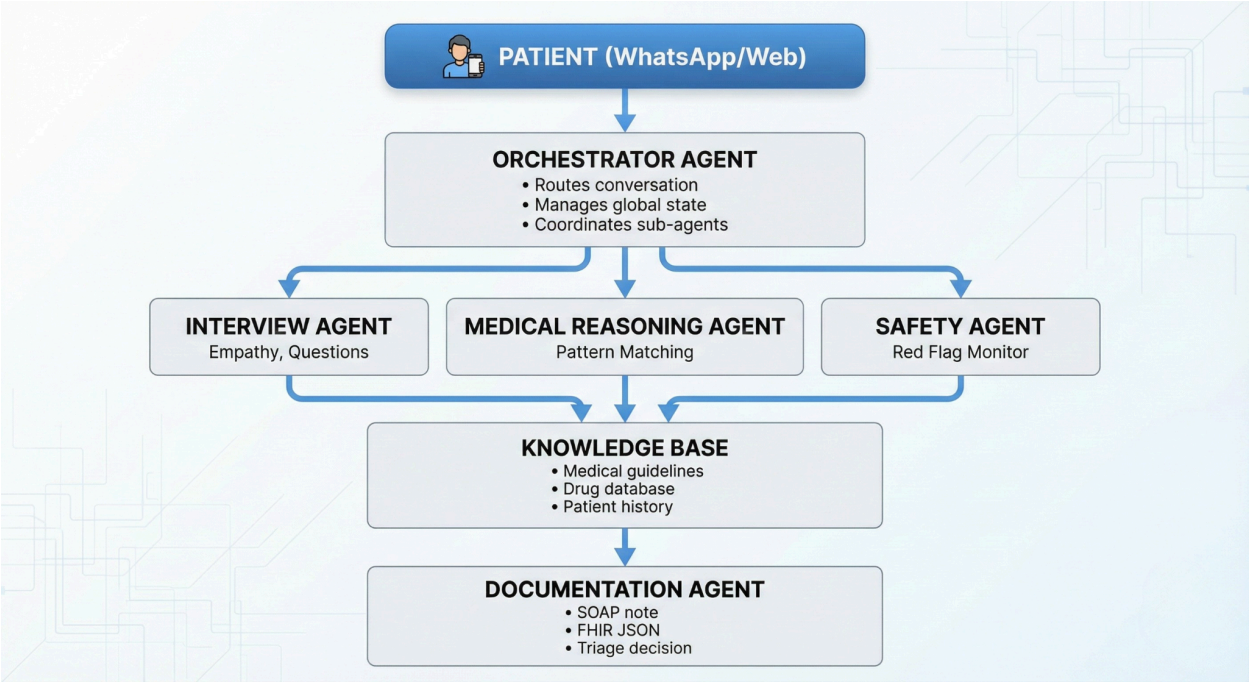
JSON


```
{
  "case_id": "DM_001",
  "chief_complaint": "تھکاوٹ اور پیاس (fatigue and thirst)",
  "language": "urdu",
  "red_flags": [
    "polyuria_with_nocturia",
    "polydipsia",
    "unintentional_weight_loss"
  ],
  "ground_truth": {
    "likely_diagnosis": "Type 2 Diabetes Mellitus",
    "triage_level": "URGENT"
  }
}
```


7. TECHNICAL ARCHITECTURE (Suggested)

Reference Multi-Agent System:

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Suggested Tech Stack

Layer	Recommended Technologies
LLM / Reasoning	GPT-4/3.5, Claude, Gemini, LLaMA, MedPaLM, BioGPT
Orchestration	LangGraph, AutoGen, CrewAI, LangChain, Semantic Kernel
Knowledge (RAG)	Pinecone, ChromaDB, Weaviate, Milvus, PubMed, WHO
Interfaces	Streamlit, Gradio, React, Flutter, WhatsApp API, Whisper

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Storage	PostgreSQL, MongoDB, Firebase, AWS S3, Pinecone
Security	HIPAA, TLS, OAuth2, Encryption, Audit Logs
Deployment	Vercel, Docker, Kubernetes, AWS Lambda, GitHub Actions

8. JUDGING CRITERIA (100 POINTS)

Criteria	Points	Focus Area
Clinical Safety	35	Emergency detection, zero false negatives on red flags.
Agentic Sophistication	25	Multi-step reasoning, adaptive behavior, tool usage.
Communication Quality	20	Empathy, cultural appropriateness, language handling.
Creativity	20	How much the team goes with the creativity.

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10. TIMELINE & PRIZES

- **Grand Prize:** 30,000 PKR + Pilot opportunity.
- **Runner-Up / Third Place:** 15,000 PKR.
- **Special Prizes:** Best Multilingual System, Most Innovative Interface.

11. FAQ

- **Do I need medical knowledge?** No, mentors and datasets are provided.
- **Can I work alone?** No, teams of 2–4 are required for diversity of skill.
- **What if I don't finish?** Depth > Breadth. One perfect scenario is better than five broken ones.

Believe in your team, trust your creativity, and enjoy the journey.
Good luck Champions!

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